An Eight-Term Counterexample to the Special Image Conjecture in Three Pairs

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Abstract

We give an explicit counterexample to the Special Image Conjecture in three contraction pairs. In $\mathbb{C}[\tau, w, v, t, z, y]$, let

$$f = \tau^3(t+z)\bigl(wt^3 - vy(t+y)^2\bigr), \qquad g = z.$$

For the standard contraction map $\mathcal{E}$, we prove for every $m \geq 1$ that

$$\mathcal{E}(f^m) = 0, \qquad [t]\mathcal{E}(gf^m) = (3m+1)!\, m!.$$

Thus every positive power of f lies in the image of the operators $\partial_t - \tau, \partial_z - w, \partial_y - v$, whereas $gf^m \notin \mathcal{M}_3$ for every $m \geq 1$. The polynomial f has eight terms and the multiplier is linear. Consequently the minimum number of pairs in which the Special Image Conjecture can fail is either two or three.

1 The counterexample

For $r \geq 1$, write

$$R_r = \mathbb{C}[\zeta_1, \ldots, \zeta_r, z_1, \ldots, z_r], \qquad \mathcal{M}_r = \sum_{i=1}^{r} (\partial_{z_i} - \zeta_i) R_r.$$

The Special Image Conjecture asserts that $\mathcal{M}_r$ is a Mathieu subspace: if $h^m \in \mathcal{M}_r$ for every $m \geq 1$, then $qh^m \in \mathcal{M}_r$ for every $q \in R_r$ and all sufficiently large m; see Zhao [1] and van den Essen, Wright, and Zhao [2].

Define the polynomial-valued contraction map

$$\mathcal{E}_r : R_r \longrightarrow \mathbb{C}[z_1, \ldots, z_r], \qquad \mathcal{E}_r(\zeta^\alpha z^\beta) = \partial_z^\alpha z^\beta.$$

Lemma 1. *For every $r \geq 1$, one has $\mathcal{M}_r = \ker \mathcal{E}_r$.*

Proof. For $h \in R_r$,

$$\mathcal{E}_r(\partial_{z_i} h) = \partial_{z_i} \mathcal{E}_r(h) = \mathcal{E}_r(\zeta_i h).$$

Thus $\mathcal{E}_r((\partial_{z_i} - \zeta_i)h) = 0$, so $\mathcal{M}_r \subseteq \ker \mathcal{E}_r$. Conversely,

$$\zeta_i h \equiv \partial_{z_i} h \pmod{\mathcal{M}_r}.$$

Repeatedly applying this congruence gives, for every monomial,

$$\zeta^\alpha z^\beta \equiv \partial_z^\alpha z^\beta = \mathcal{E}_r(\zeta^\alpha z^\beta) \pmod{\mathcal{M}_r}.$$

By linearity, $p \equiv \mathcal{E}_r(p) \pmod{\mathcal{M}_r}$. Since $\mathcal{E}_r$ is the identity on $\mathbb{C}[z_1, \ldots, z_r]$, $\mathcal{E}_r(p) = 0$ implies $p \in \mathcal{M}_r$. $\square$

Use the pairs

$$(\tau,t),\qquad (w,z),\qquad (v,y)$$

and put

$$\mathcal{M}_3=(\partial_t-\tau)R_3+(\partial_z-w)R_3+(\partial_y-v)R_3.$$

Theorem 2. *Let*

$$\boxed{f=\tau^3(t+z)\big(wt^3-vy(t+y)^2\big),\qquad g=z.}$$

Then, for every $m\ge 1$,

$$\mathcal{E}_3(f^m)=0,\qquad [t]\mathcal{E}_3(gf^m)=(3m+1)!\,m!\neq 0.$$

In particular, $\mathcal{E}_3(gf^m)\neq 0$. Consequently $f^m\in\mathcal{M}_3$ and $gf^m\notin\mathcal{M}_3$ for every $m\ge 1$.

In expanded form,

$$\boxed{f=\tau^3\big(wt^4+wt^3z-vt^3y-2vt^2y^2-vty^3-vt^2yz-2vty^2z-vy^3z\big).}$$

This displays the advertised eight terms directly.

2 Proof

First omit the pair (τ,t). On $\mathbb{C}[W,Z,V,Y]$, define

$$\mathcal{F}(W^aZ^bV^cY^d)=\delta_{a,b}\delta_{c,d}\,a!\,c!$$

and set

$$P=(1+Z)\big(W-VY(1+Y)^2\big).$$

Expanding according to the number k of factors chosen from the second summand gives

$$P^m=\sum_{k=0}^{m}(-1)^k\binom{m}{k}W^{m-k}V^k(1+Z)^mY^k(1+Y)^{2k}.$$

For a nonzero contraction, the W,Z pair selects the coefficient of Z^{m-k}, while the V,Y pair selects the constant term of $(1+Y)^{2k}$. Hence

$$\mathcal{F}(P^m)=m!\sum_{k=0}^{m}(-1)^k\binom{m}{k}=0. \tag{1}$$

After multiplication by Z, the first pair instead selects $[Z^{m-k-1}](1+Z)^m=\binom{m}{k+1}$. Therefore

$$\mathcal{F}(ZP^m)=m!\sum_{k=0}^{m-1}(-1)^k\binom{m}{k+1}=m!. \tag{2}$$

Homogenize in the coordinate variables Z,Y, treating W,V as coefficient, or dual, variables:

$$\widetilde{P}(t,w,z,v,y)=t^4P\left(w,\frac{z}{t},v,\frac{y}{t}\right)=(t+z)\big(wt^3-vy(t+y)^2\big).$$

Each factor of P has total W,V-degree one. Every monomial of P^m that survives $\mathcal{F}$ therefore has total (Z,Y)-degree m, matching its total W,V-degree. Its coordinate homogenization contains

$t^{4m-m} = t^{3m}$, exactly matching the power τ^{3m}. The new contraction contributes $(3m)!$, and (1) gives

$$\mathcal{E}_3(f^m) = (3m)!\mathcal{F}(P^m) = 0.$$

For ZP^m, a surviving term of P^m has total (Z, Y)-degree $m-1$. Its homogenization contains $t^{4m-(m-1)} = t^{3m+1}$. Contracting against τ^{3m} leaves one power of t and contributes $(3m+1)!$. By (2),

$$[t]\mathcal{E}_3(zf^m) = (3m+1)!\mathcal{F}(ZP^m) = (3m+1)!\, m! \neq 0.$$

Lemma 1 completes the proof of Theorem 2.

Corollary 3. *If r_{SIC} denotes the minimum number of contraction pairs in which the Special Image Conjecture fails, then*

$$2 \leq r_{\mathrm{SIC}} \leq 3.$$

Proof. The upper bound is Theorem 2. The one-pair case follows from [2, Theorem 2.8]: take the $\mathbb{Q}$-algebra $A = \mathbb{C}[\tau]$ and $a = \tau$. Then a is a non-zero-divisor and the principal ideal $Aa = (\tau)$ is radical, precisely the hypotheses of that theorem. □

3 Size, scope, and verification

The coefficients of f lie in $\{1, -1, -2\}$. With respect to (τ, w, v)-degree and (t, z, y)-degree, f has bidegree $(4, 4)$ and ordinary degree eight; $g = z$ has bidegree $(0, 1)$. The polynomial f is not of the separated form $\lambda(\zeta)P(z)$; consequently the example does not give a three-variable Generalized Vanishing Conjecture counterexample.

Thus the known one-pair result and Theorem 2 give

$$2 \leq r_{\mathrm{SIC}} \leq 3.$$

We make no claim here concerning the two-pair case or absolute literature priority.

The all-order proof above is independent of computation. An exact sparse-arithmetic checker expands f, verifies the contractions through $m = 10$, and checks the two binomial identities through $m = 99$. The checker is included in the accompanying Zenodo record and in the companion repository.[1] Its SHA-256 digest in this version is

`ac5408c1278e6fe2d3f78b408bd2165454833db42a669cd929570ba1f2e30d5b.`

As a low-order checksum,

$$\mathcal{E}_3(f) = 0, \qquad \mathcal{E}_3(zf) = 24t + 6z.$$

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References

[1] W. Zhao, *Images of commuting differential operators of order one with constant leading coefficients*, J. Algebra **324** (2010), 231–247. doi:10.1016/j.jalgebra.2010.04.022.

[2] A. van den Essen, D. Wright, and W. Zhao, *On the Image Conjecture*, J. Algebra **340** (2011), 211–224. doi:10.1016/j.jalgebra.2011.04.036.

[1] Repository path: `scripts/verify_three_pair_image_mathieu_counterexample.py`.